

BHARATHI K | AI ENGINEER

📍 Chennai, India | 📞 +91 8248754956 | ✉ bharathinathan31@gmail.com | [🌐 LinkedIn Profile](#) | [🐙 GitHub](#) | [📁 Portfolio](#)

SUMMARY

- AI Engineer with hands-on experience in building LLM-based conversational AI systems integrating structured databases and unstructured documents.
- Experienced in SQL generation pipelines, Retrieval-Augmented Generation (RAG), vector search, and prompt engineering using Azure Llama models..
- Built end-to-end AI pipelines enabling natural language querying over databases, PDFs, and Excel data, delivering automated insights, summaries, and recommendations.
- Strong foundation in Python and SQL with real-world experience in AI system development, optimization, and intern mentoring.

EXPERIENCE

Prowesstics IT Services Pvt Ltd, Chennai

Junior Software Engineer

June 2025 – Present

GenAI / AI Projects:

- Took ownership of key modules in an LLM-based conversational AI system for natural language database querying using Azure Models.
- Building and maintaining SQL generation pipelines using schema-based retrieval and prompt engineering techniques.
- Implemented and maintained RAG-based document processing for PDF and Excel data using embeddings and vector search (FAISS).
- Built multi-step AI workflows including query processing, retrieval, SQL validation, error handling, and response generation.
- Integrated vector databases and semantic search for improved context retrieval and response accuracy.
- Contributed to end-to-end AI pipeline including data ingestion, processing, chart generation, and business insight delivery.
- Trained interns on AI pipeline components, system architecture, and workflow processes.

Data & Analytics Projects:

- Contributed to NIFCO project — connected Oracle DB to Power BI, configured gateway and scheduled refresh settings.
- Developed interactive dashboards using DAX functions and published reports to Power BI Service with workspace management.
- Performed data cleaning, SQL queries, and data modeling to support reporting and analytics needs.
- Assisted in deep learning model training for Gender and Age Classification (ProSight project — synthetic dataset creation and evaluation).

PROJECTS

AI Chatbot with RAG, SQL & Document Intelligence

Prowesstics IT Services Pvt Ltd

Azure OpenAI, RAG, FAISS, Python, SQL, LangChain

- Contributed to building an end-to-end AI chatbot integrating SQL generation and RAG-based document retrieval using Azure Models.
- Took ownership of PDF and Excel ingestion pipeline — implemented chunking, embedding, and contextual response generation.
- Implemented and maintained vector search using FAISS and embedding models for semantic retrieval over unstructured documents.
- Built LLM-powered SQL generation, automated chart creation, business summaries, and data-driven recommendations.
- Reduced query time and improved retrieval accuracy

NIFCO – Power BI & Oracle Database Integration (POC)

Prowesstics IT Services Pvt Ltd

Power BI, Oracle DB, PostgreSQL, DAX, Power Query, Power BI Service

- Connected Oracle DB to Power BI and configured gateway, credentials, and scheduled refresh settings.
- Developed interactive dashboard using DAX functions; published to Power BI Service with workspace and access management.
- Collaborated with team to deliver actionable, data-driven insights through scalable reporting solutions.

ProSight – Customer Journey Analytics (POC)

Prowesstics IT Services Pvt Ltd

Python, Deep Learning, Synthetic Data Generation

- Developed to high-level project architecture and documentation; conducted market and competitor research.
- Created synthetic datasets simulating customer footfall and behavior; Implemented in DL model training for Gender and Age Classification.

SKILLS

AI / GenAI: LLM, RAG, Prompt Engineering, LangChain, LangGraph, Vector Databases (CHROMA & FAISS), Azure Models.

Programming & DB: Python, MySQL, PostgreSQL, Oracle Database

Data & Analytics: Power BI (Desktop / Service), Advanced Excel, Tableau, DAX

Other: Deep Learning Concepts, Git, ClickUp, Google Sheets, Gitlab (Basic)

CERTIFICATIONS

- Data Analytics Internship — Besant Technologies | Certificate Link – [Click Here](#)
- Data Analytics with AI — Workshop, Dr. M.G.R Educational and Research Institute | Certificate Link – [Click Here](#)
- Programming in Python — Swayam (NPTEL) | Certificate Link – [Click Here](#)

EDUCATION

Master of Computer Application (MCA)

2022 – 2024

Dr. M.G.R Educational and Research Institute | 75%

Bachelor of Computer Application (BCA)

2019 – 2022

St. Thomas College of Arts and Science | 70%